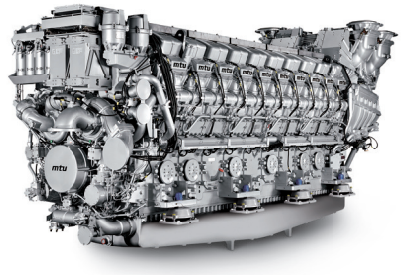
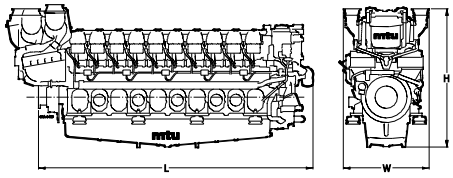
Marine

DIESEL ENGINE 20V 8000 M91L

for fast vessels with low load factors (1DS)



Engine	Dimensions (L x W x H) mm (in)	Mass, dry kg (lbs)
M91L	6645 x 2040 x 3375 (261.6 x 80.3 x 132.9)	49600 (109348)
		Mass, wet kg (lbs)
		51965 (114563)



Typical applications: OPVs, corvettes, frigates, destroyers and landing ships with naval requirement and large yachts with high acoustic requirements respectively

Optional equipment and finishing shown. Standard may vary.

Engine type		20V 8000 M91L
Rated power ICFN	kW	10000
	(bhp)	(13410)
Speed	rpm	1008
No. of cylinders		20
Bore/stroke	mm (in)	265/315 (10.4/12.4)
Displacement	l (cu in)	347.4 (21200)
Optimization of exhaust emissions*		IMO II

* IMO - International Maritime Organisation (MARPOL)

A Rolls-Royce solution

Fuel consumption *		20V 8000 M91L
at rated power	g/kWh	199
	l/hr	2397,6
	gal/hr	633.3

* Tolerance +5% per ISO 3046, Diesel fuel to DIN EN 590 with a min L.H.V. of 42800kJ/kg (18390 BTU/lb)

Standard equipment	
Starting system	Air starter motor, 15 bar; press. reduct. station 40/15 bar, coolant preheating system
Oil system	Lube oil pump, automatic filter with backflushing, centrifugal oil filter, lube-oil heat exchanger, lube oil priming pump, lube oil level monitoring/replenishment system, switchboxes for lube oil replenishment and priming pumps
Fuel system	Fuel delivery pump, fuel duplex filter with diverter valve, “common rail” fuel injection system with high-pressure pump, pressure accumulator and electronically fuel injection with cylinder cutout system, jacketed HP fuel lines, leak-off fuel tank level monitored, fuel hand pump, fuel pre-filter with water separator, fuel recooler
Cooling system	MTU-split-circuit coolant system, coolant-to-raw water plate core heat exchanger, centrifugal raw water pump with priming system, coolant circulation pump, coolant expansion tank
Combustion air system	Engine coolant temperature-controlled intercooler, sequential turbocharging with 4 water-cooled turbochargers, on-engine set of combustion-air filters
Exhaust system	On-engine exhaust manifolds, exhaust bellow
Mounting system	Highly resilient mounts for high acoustic requirements and/or shock requirements to NATO Standards
Power transmission	Torsional and offset compensating couplings
Engine management system	Engine control and monitoring system (MDEC), interface to remote control and monitoring system, local operating panel (LOP)
Interfaces	Flexible joints (hose lines, rubber bellows)

Optional equipment	
Starting system	Compressed air tanks
Monitoring/control system	Monitoring and control system MCS-5, remote control system RCS-5
Gearbox option	Various gearbox models
Classification	Restricted service

Reference conditions:
> Power definition according ISO 3046
> Intake air temperature 25°C/Sea water temperature 25°C
> Intake air depression 15 mbar/Exhaust back pressure 30 mbar
> Barometric pressure 1000 mbar
> Power reduction at 45°C/32°C: none

Specifications are subject to change without notice.
All dimensions are approximate, for complete information refer to installations drawing. For further information consult your MTU distributor/dealer.

Subject to change. | 3236371 | Edition 01/20 | ESM 2020-09.

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